

```
#question 1
x=int(input())
y=int(input())
if x>0:
    if y>0:
        print('제1사분면')
    else:
        print('제4사분면')
else:
    if y>0:
        print('제2사분면')
    else:
        print('제3사분면')
```

```
-10
10
제2사분면
```

```
# question 2
Y, M =map(int,input().split())
if Y%4==0 and Y%100!=0 or Y%400==0:
    if M==2:
        print(29)
    elif M in [1,3,5,7,8,10,12]:
        print(31)
    else:
        print(30)
else:
    if M==2:
        print(28)
    elif M in [1,3,5,7,8,10,12]:
        print(31)
    else:
        print(30)
```

```
2000 10
31
```

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#question 3
a,b=map(int, input().split())
sum=0
words=''
for i in range(a,b+1):
    if i%2==0:
        sum-=i
        words+='-'+str(i)
    else:
        sum+=i
        words+=''+str(i)
print(sum)
if words[0]=='':
    words=words[1:]
print(words+'='+str(sum))
```

```
5 10
-3
5-6+7-8+9-10=-3
```

```
# question 4
pose=['rock','paper','scissors']
C_ch=random.choice(pose)
P_ch=input('user choice: ')
print(f'User: {P_ch}, Computer: {C_ch}')
if P_ch==C_ch:
    print('Draw')
elif (P_ch=='rock' and C_ch=='scissors') or (P_ch=='paper' and C_ch=='rock') or (P_ch=='scissors' and C_ch=='paper'):
    print('User wins')
else:
    print('Computer wins')
```

```
user choice: scissors
User: scissors, Computer: paper
User wins
```

```
# question 5
LST=[]
while True:
    name=input('if you want to exit, input exit. input animal name: ')
    if name=='exit':
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        break
    elif name in LST:
        print('Already in the list')
    elif name not in LST:
        if len(name)<3:
            print('Name is too short')
        elif name.isalpha()==False:
            print('Name is not valid')
        else:
            LST.append(name)
LST.sort(key=len)
print(LST)

```

```

if you want to exit, input exit. input animal name: snake
if you want to exit, input exit. input animal name: tiger
if you want to exit, input exit. input animal name: elephant
if you want to exit, input exit. input animal name: crocodile
if you want to exit, input exit. input animal name: exit
['snake', 'tiger', 'elephant', 'crocodile']

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# question 6
import random
C_num=random.randint(10,99)
i=1
while True:
    P_num=str(input('Chance ' + str(i) + ': '))
    if P_num.isdigit()==False or len(P_num)!=2:
        print("This is not valid.")
    elif int(P_num)<C_num:
        print("Too low.")
        i+=1
    elif int(P_num)>C_num:
        print("Too high.")
        i+=1
    else:
        print("Correct!")
        print(f"It took you {i} chances.")
        break

```

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Chance 1: 12
Too low.
Chance 2: 66
Too high.
Chance 3: 54
Too low.
Chance 4: 58
Too high.
Chance 5: 56
Too high.
Chance 6: 55
Correct!
It took you 6 chances.

```

```

# question 7
list=[[0,0,0,0,0,0,0,0,0,0],
      [0,0,0,1,1,1,1,0,0,0],
      [0,1,1,0,0,0,0,1,1,0],
      [1,0,0,0,1,1,0,0,0,1],
      [0,1,1,0,0,0,0,1,1,0],
      [0,0,0,1,1,1,1,0,0,0]]
for y in range(len(list)):
    for x in range(len(list[y])):
        if list[y][x]==0:
            print('■',end='')
        else:print('□',end='')
    print()

```

```

■ ■ ■ ■ ■ ■ ■ ■ ■ ■
■ ■ ■ □ □ □ ■ ■ ■ ■
■ □ □ ■ ■ ■ ■ □ □ ■
□ ■ ■ ■ □ □ ■ ■ ■ □
■ □ □ ■ ■ ■ ■ □ □ ■
■ ■ ■ □ □ □ □ ■ ■ ■

```

```

# question 8,9
import csv, string, random
def name():
    fir='김 이 박 최 정 황 강 백 성 오 신 안'.split(' ')
    sec='태 명 동 현 지 유 운 장 준 은 선 주'.split(' ')
    las='찬 민 원 현 흥 연 현 우 영 대 혁 진'.split(' ')
    name=random.choice(fir)+random.choice(sec)+random.choice(las)
    return name
def phone():

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num='010-'
for k in range(8):
    num+=str(random.randint(0,9))
    if k==3:
        num+="-"
return num
def mail():
    mane=""
    lag=string.ascii_letters
    num=string.digits
    for q in range(8):
        mane+=str(random.choice(lag+num))
    return mane+"@gmail.com"
filename='MMMMM.csv'
with open(filename,'w',newline='',encoding='cp949') as f:
    Writer=csv.writer(f)
    Writer.writerow(['Name','Phone','MAil'])
    for i in range(20):
        Writer.writerow([name(),phone(),mail()])

```

```

# question 10,11,12
Lt=[12,314,23,22,7,88,5,0,99,65]
def bubble_sort(list):
    for i in range(len(list)-1):
        for j in range(len(list)-1-i):
            if list[j]>list[j+1]:
                list[j],list[j+1]=list[j+1],list[j]
        print(list)

def improved_bubble_sort(list):
    for i in range(len(list)-1):
        Flag=True
        for j in range(len(list)-1-i):
            if list[j]>list[j+1]:
                Flag=False
                list[j],list[j+1]=list[j+1],list[j]
        if Flag==True:
            print('COMPLETE')
            break
    print(list)

improved_bubble_sort(Lt)

```

```

[12, 23, 22, 7, 88, 5, 0, 99, 65, 314]
[12, 22, 7, 23, 5, 0, 88, 65, 99, 314]
[12, 7, 22, 5, 0, 23, 65, 88, 99, 314]
[7, 12, 5, 0, 22, 23, 65, 88, 99, 314]
[7, 5, 0, 12, 22, 23, 65, 88, 99, 314]
[5, 0, 7, 12, 22, 23, 65, 88, 99, 314]
[0, 5, 7, 12, 22, 23, 65, 88, 99, 314]
COMPLETE

```

```

Lt=[12,314,23,22,7,88,5,0,99,65]
def selection_sort(list):
    for i in range(len(list)-1):
        min=i
        for j in range(i+1,len(list)):
            if list[j]<list[min]:
                min=j
        list[i],list[min]=list[min],list[i]
        print(list)

selection_sort(Lt)

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[0, 314, 23, 22, 7, 88, 5, 12, 99, 65]
[0, 5, 23, 22, 7, 88, 314, 12, 99, 65]
[0, 5, 7, 22, 23, 88, 314, 12, 99, 65]
[0, 5, 7, 12, 23, 88, 314, 22, 99, 65]
[0, 5, 7, 12, 22, 88, 314, 23, 99, 65]
[0, 5, 7, 12, 22, 23, 314, 88, 99, 65]
[0, 5, 7, 12, 22, 23, 65, 88, 99, 314]
[0, 5, 7, 12, 22, 23, 65, 88, 99, 314]
[0, 5, 7, 12, 22, 23, 65, 88, 99, 314]

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```

Lt=[10,21,33,333,42,7,45,88,86,94,121,52,73,39]
def insertion_sort(list):
    for i in range(1,len(list)):
        key=list[i]
        j=i-1

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while j>=0 and list[j]>key:
    list[j+1]=list[j]
    j-=1
list[j+1]=key
print(list)
insertion_sort(Lt)
```

```
[10, 21, 33, 333, 42, 7, 45, 88, 86, 94, 121, 52, 73, 39]
[10, 21, 33, 333, 42, 7, 45, 88, 86, 94, 121, 52, 73, 39]
[10, 21, 33, 333, 42, 7, 45, 88, 86, 94, 121, 52, 73, 39]
[10, 21, 33, 42, 333, 7, 45, 88, 86, 94, 121, 52, 73, 39]
[7, 10, 21, 33, 42, 333, 45, 88, 86, 94, 121, 52, 73, 39]
[7, 10, 21, 33, 42, 45, 333, 88, 86, 94, 121, 52, 73, 39]
[7, 10, 21, 33, 42, 45, 88, 333, 86, 94, 121, 52, 73, 39]
[7, 10, 21, 33, 42, 45, 86, 88, 333, 94, 121, 52, 73, 39]
[7, 10, 21, 33, 42, 45, 86, 88, 94, 333, 121, 52, 73, 39]
[7, 10, 21, 33, 42, 45, 86, 88, 94, 121, 333, 52, 73, 39]
[7, 10, 21, 33, 42, 45, 52, 86, 88, 94, 121, 333, 73, 39]
[7, 10, 21, 33, 42, 45, 52, 73, 86, 88, 94, 121, 333, 39]
[7, 10, 21, 33, 39, 42, 45, 52, 73, 86, 88, 94, 121, 333]
```